

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 07 · Transformations of Functions

With Answers

Questions with full worked solutions

- 24 pages
- Past papers up to June 2026

Transformations of Functions

Your classified route

7
questions

55
marks

3
families

6
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Transformations of Functions

Families and variants

F01 • Coordinate maps under function notation

V01 Translations and reflections of key points and asymptotes

V02 Horizontal input scale/reflection with equations and inequalities

V03 Combined input transformation and intercept/multiplicity mapping

F02 • Transformed algebraic graphs and exact intersections

V01 Square-root transformations with exact surd intersection

V02 Translated reciprocal graph followed by line intersection

F03 • Transformations on trigonometric graphs

V01 Identify a sine graph and transform a labelled extremum

DR ESLAM AHMED

Coordinate maps under function notation

WHAT THIS FAMILY ASKS

Transform an already-known graph and carry its labelled features through the coordinate map.

VARIANT ROUTE

V01 Translations and reflections of key points and asymptotes

V02 Horizontal input scale/reflection with equations and inequalities

V03 Combined input transformation and intercept/multiplicity mapping

3 Complete Classified questions

20 total marks

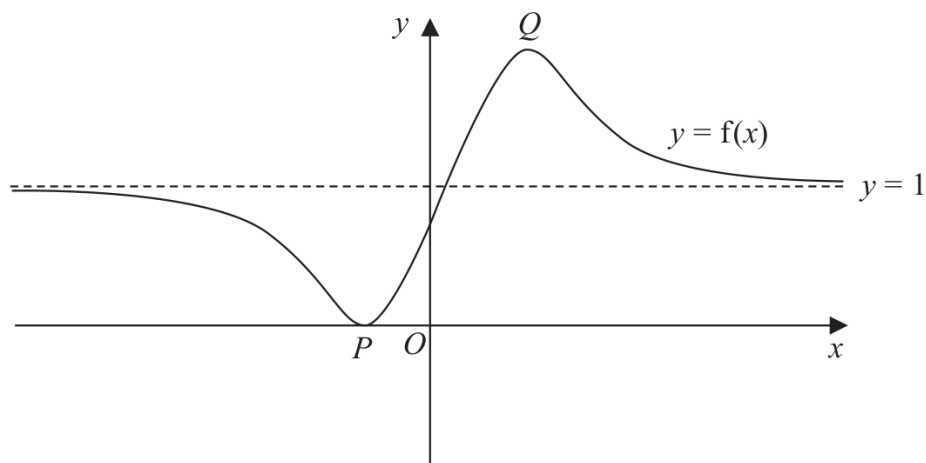


Figure 1

Figure 1 shows a sketch of a curve with equation $y = f(x)$

The curve has a minimum at $P(-1, 0)$ and a maximum at $Q\left(\frac{3}{2}, 2\right)$

The line with equation $y = 1$ is the only asymptote to the curve.

On separate diagrams sketch the curves with equation

(i) $y = f(x) - 2$ (3)

(ii) $y = f(-x)$ (3)

On each sketch you must clearly state

- the coordinates of the maximum and minimum points
- the equation of the asymptote

(Total 6 marks)

Question 1

WMA11/01 May/June 2022

T07 Transformations of Functions

F01 Coordinate maps under function notation

V01 Translations and reflections of key points and asymptotes

1

Translate for $y = f(x) - 2$

Subtracting 2 moves every point of the graph down by 2 units.

2

State the new key featuresThe minimum $P(-1, 0)$ becomes $(-1, -2)$, the maximum $Q(\frac{3}{2}, 2)$ becomes $(\frac{3}{2}, 0)$, and the asymptote $y = 1$ becomes $y = -1$.

3

Reflect for $y = f(-x)$ Replacing x by $-x$ reflects the graph in the y-axis.

4

State the reflected key featuresThe minimum $P(-1, 0)$ becomes $(1, 0)$, the maximum $Q(\frac{3}{2}, 2)$ becomes $(-\frac{3}{2}, 2)$, and the horizontal asymptote remains $y = 1$.**FINAL ANSWER**

(i) minimum $(-1, -2)$, maximum $(\frac{3}{2}, 0)$, asymptote $y = -1$. (ii) minimum $(1, 0)$, maximum $(-\frac{3}{2}, 2)$, asymptote $y = 1$.

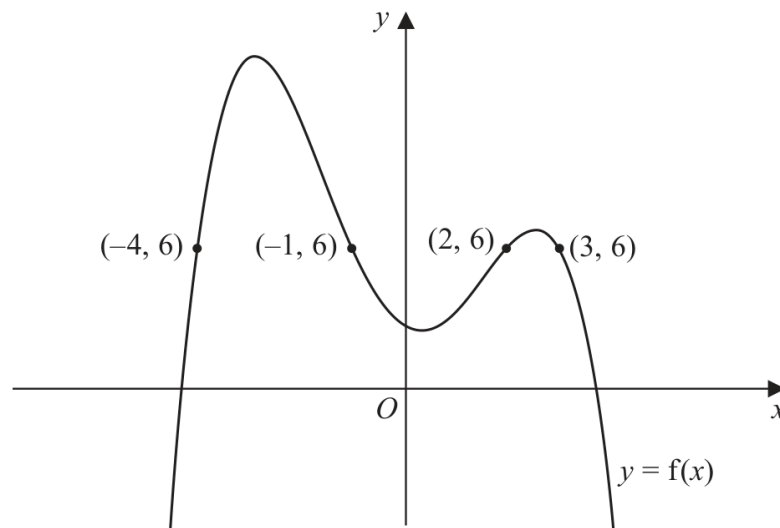


Figure 1

Figure 1 shows the curve with equation $y = f(x)$.

The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

(a) Using Figure 1, find the range of values of x for which

$$f(x) < 6$$

(3)

(b) State the largest solution of the equation

$$f(2x) = 6$$

(1)

(c) (i) Sketch the curve with equation $y = f(-x)$.

On your sketch, state the coordinates of the points to which P , Q , R and S are transformed.

(ii) Hence find the set of values of x for which

$$f(-x) \geq 6 \text{ and } x < 0$$

(4)

(Total for Question 7 is 8 marks)

Question 2

WMA11/01 October 2022

T07 Transformations of Functions

F01 Coordinate maps under function notation

V02 Horizontal input scale/reflection with equations and inequalities

1

Read $f(x) < 6$ from the graphThe curve is below the line $y = 6$ for $x < -4$, for $-1 < x < 2$, and for $x > 3$.

2

Solve $f(2x) = 6$ Since $f(u) = 6$ at $u = -4, -1, 2, 3$, we set $2x = -4, -1, 2, 3$. The solutions are $x = -2, -\frac{1}{2}, 1, \frac{3}{2}$.

3

State the largest solutionThe largest solution is $x = \frac{3}{2}$.

4

Reflect the points in the y-axisFor $y = f(-x)$, each point (x, y) becomes $(-x, y)$.

5

List the transformed points $P(-4, 6) \rightarrow (4, 6)$, $Q(-1, 6) \rightarrow (1, 6)$, $R(2, 6) \rightarrow (-2, 6)$, and $S(3, 6) \rightarrow (-3, 6)$.

6

Use the reflected graphFor $x < 0$, the part of the reflected graph on or above $y = 6$ is from $x = -3$ to $x = -2$.

FINAL ANSWER

(a) $x < -4$, $-1 < x < 2$, $x > 3$. (b) $\frac{3}{2}$. (c)(ii) $-3 \leq x \leq -2$.

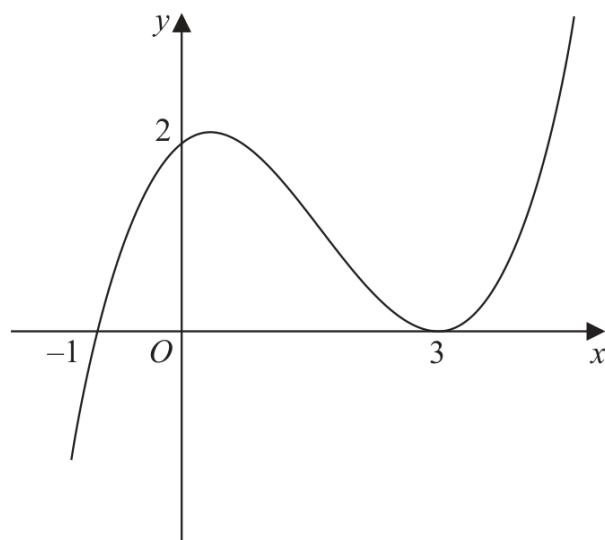


Figure 1

Figure 1 shows a sketch of the curve with equation $y = f(x)$.

The curve passes through the points $(-1, 0)$ and $(0, 2)$ and touches the x -axis at the point $(3, 0)$.

On separate diagrams, sketch the curve with equation

(a) $y = f(x + 3)$

(b) $y = f(-3x)$

On each diagram, show clearly the coordinates of all the points where the curve cuts or touches the coordinate axes.

(Total for Question 3 is 6 marks)

Question 3

WMA11/01 May/June 2024

T07 Transformations of Functions

F01 Coordinate maps under function notation

V03 Combined input transformation and intercept/multiplicity mapping

1

Read the original key points

The original curve passes through $(-1, 0)$ and $(0, 2)$, and touches the x-axis at $(3, 0)$.

2

Translate the graph left

$y = f(x + 3)$ is a translation 3 units left. So $(-1, 0)$ moves to $(-4, 0)$, $(0, 2)$ moves to $(-3, 2)$, and $(3, 0)$ moves to $(0, 0)$.

3

State the axis intercepts for part a

The curve cuts the x-axis at $(-4, 0)$ and touches it at $(0, 0)$. The y-intercept is also $(0, 0)$.

4

Transform y equals f of negative 3x

For $y = f(-3x)$, a point with original x-coordinate X moves to $x = -\frac{X}{3}$.

5

State the axis intercepts for part b

$(-1, 0)$ moves to $(\frac{1}{3}, 0)$, $(3, 0)$ moves to $(-1, 0)$, and $(0, 2)$ stays at $(0, 2)$.

FINAL ANSWER

(a) intercepts $(-4, 0)$ and $(0, 0)$, with $(0, 0)$ a touch point. (b) intercepts $(-1, 0)$, $(\frac{1}{3}, 0)$ and $(0, 2)$, with $(-1, 0)$ a touch point.

Transformed algebraic graphs and exact intersections

WHAT THIS FAMILY ASKS

Describe the graph transformations of a named algebraic function and then solve exactly where two transformed versions meet.

VARIANT ROUTE

V01 Square-root transformations with exact surd intersection

V02 Translated reciprocal graph followed by line intersection

3 Complete Classified questions

29 total marks

9.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

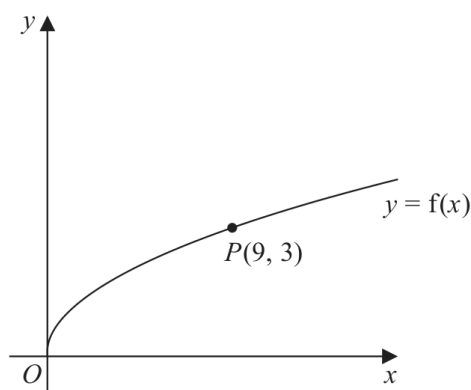


Figure 5

Figure 5 shows a sketch of the curve with equation $y = f(x)$ where

$$f(x) = \sqrt{x} \quad x > 0$$

The point $P(9, 3)$ lies on the curve and is shown in Figure 5.

On the next page there is a copy of Figure 5 called Diagram 1.

(a) On Diagram 1, sketch and clearly label the graphs of

$$y = f(2x) \quad \text{and} \quad y = f(x) + 3$$

Show on each graph the coordinates of the point to which P is transformed.

(3)

The graph of $y = f(2x)$ meets the graph of $y = f(x) + 3$ at the point Q .

(b) Show that the x coordinate of Q is the solution of

$$\sqrt{x} = 3(\sqrt{2} + 1)$$

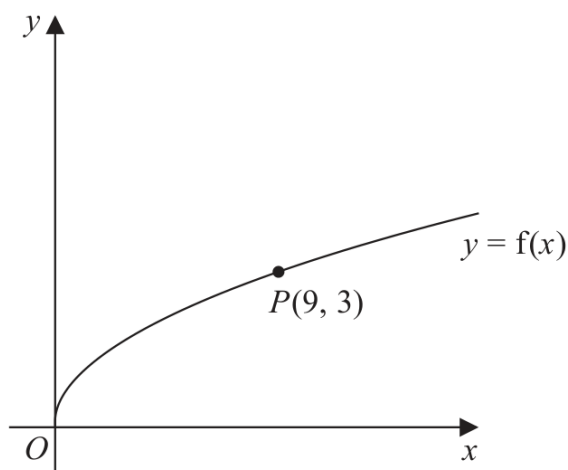
(3)

(c) Hence find, in simplest form, the coordinates of Q .

(3)

QUESTION CONTINUES

Required answer objects follow.

**Diagram 1**

Turn over for a copy of Diagram 1 if you need to redraw your graphs.

(Total 9 marks)

Question 4

WMA11/01 October 2021

T07 Transformations of Functions

F02 Transformed algebraic graphs and exact intersections

V01 Square-root transformations with exact surd intersection

1

Transform P on $y = f(2x)$ $y = f(2x)$ is a horizontal scale factor of $\frac{1}{2}$, so $P(9, 3)$ maps to $\left(\frac{9}{2}, 3\right)$.

2

Translate point P $y = f(x) + 3$ is a translation 3 units upward, so $P(9, 3)$ maps to $(9, 6)$.

3

Set up the intersection equationAt Q, $f(2x) = f(x) + 3$. Since $f(x) = \sqrt{x}$, this gives $\sqrt{2x} = \sqrt{x} + 3$.

4

Show the required equationLet $s = \sqrt{x}$. Then $\sqrt{2x} = \sqrt{2}s$, so $\sqrt{2}s = s + 3$. Hence $(\sqrt{2} - 1)s = 3$ and $s = \frac{3}{\sqrt{2} - 1} = 3(\sqrt{2} + 1)$.

5

Find the x-coordinate $\sqrt{x} = 3(\sqrt{2} + 1)$, so $x = 9(\sqrt{2} + 1)^2 = 9(3 + 2\sqrt{2}) = 27 + 18\sqrt{2}$.

6

Find the y-coordinate $y = \sqrt{x} + 3 = 3(\sqrt{2} + 1) + 3 = 6 + 3\sqrt{2}$. Therefore $Q = (27 + 18\sqrt{2}, 6 + 3\sqrt{2})$.

FINAL ANSWER

(a) $P \mapsto \left(\frac{9}{2}, 3\right)$ on $y = f(2x)$, $P \mapsto (9, 6)$ on $y = f(x) + 3$. (c) $Q = (27 + 18\sqrt{2}, 6 + 3\sqrt{2})$.

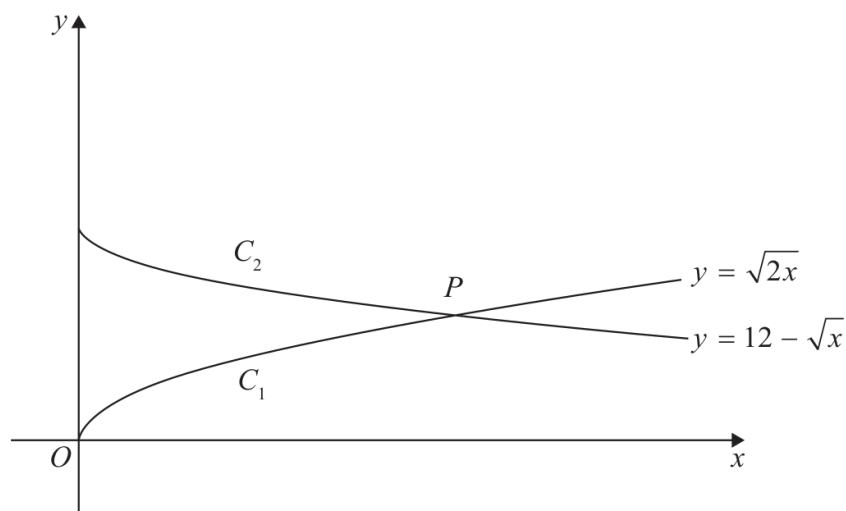


Figure 4

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Figure 4 shows a sketch of

- the graph C_1 with equation $y = \sqrt{2x}$
- the graph C_2 with equation $y = 12 - \sqrt{x}$

(a) Describe fully the single transformation that would transform

- the graph with equation $y = \sqrt{x}$ onto C_1
- the graph with equation $y = -\sqrt{x}$ onto C_2

(4)

The graphs C_1 and C_2 meet at the point P , as shown in Figure 4.

(b) (i) Show that the x coordinate of P is a solution of

$$\sqrt{x} = 12(\sqrt{2} - 1)$$

(ii) Hence find, in simplest form, the exact coordinates of P .

(6)

(Total for Question 9 is 10 marks)

Question 5

WMA11/01 May/June 2025

T07 Transformations of Functions

F02 Transformed algebraic graphs and exact intersections

V01 Square-root transformations with exact surd intersection

1

Transform y equals root x to C_1

C_1 has equation $y = \sqrt{2x}$. This is obtained from $y = \sqrt{x}$ by a stretch parallel to the x -axis with scale factor $\frac{1}{2}$.

2

Equivalent description

Equivalently, $y = \sqrt{2x} = \sqrt{2}\sqrt{x}$ is a stretch parallel to the y -axis with scale factor $\sqrt{2}$. The mark scheme also accepts the horizontal stretch description.

3

Translate the reflected curve

C_2 has equation $y = 12 - \sqrt{x} = -\sqrt{x} + 12$, so it is a translation of $y = -\sqrt{x}$ by $\begin{pmatrix} 0 \\ 12 \end{pmatrix}$.

4

Set the curves equal

At P , $\sqrt{2x} = 12 - \sqrt{x}$.

5

Factor out root x

$\sqrt{2}\sqrt{x} + \sqrt{x} = 12$, so $(\sqrt{2} + 1)\sqrt{x} = 12$.

6

Rationalise

$$\sqrt{x} = \frac{12}{\sqrt{2} + 1} = \frac{12(\sqrt{2} - 1)}{(\sqrt{2} + 1)(\sqrt{2} - 1)} = 12(\sqrt{2} - 1).$$

7

Find x

$$x = [12(\sqrt{2} - 1)]^2 = 144(3 - 2\sqrt{2}).$$

8

Find y

$$y = \sqrt{2x} = \sqrt{2}\sqrt{x} = \sqrt{2} \cdot 12(\sqrt{2} - 1) = 24 - 12\sqrt{2}.$$

FINAL ANSWER

(a)(i) **stretch parallel to the x-axis scale factor $\frac{1}{2}$ (equivalently vertical stretch scale factor $\sqrt{2}$).** (a)(ii) **translation by $\begin{pmatrix} 0 \\ 12 \end{pmatrix}$.** (b) $P = (144(3 - 2\sqrt{2}), 24 - 12\sqrt{2})$.

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7. (a) On Diagram 1, sketch a graph of the curve C with equation

$$y = \frac{6}{x} \quad x \neq 0 \quad (2)$$

The curve C is transformed onto the curve with equation $y = \frac{6}{x-2} \quad x \neq 2$

- (b) Fully describe this transformation. (2)

The curve with equation

$$y = \frac{6}{x-2} \quad x \neq 2$$

and the line with equation

$$y = kx + 7 \quad \text{where } k \text{ is a constant}$$

intersect at exactly two points, P and Q .

Given that the x coordinate of point P is -4

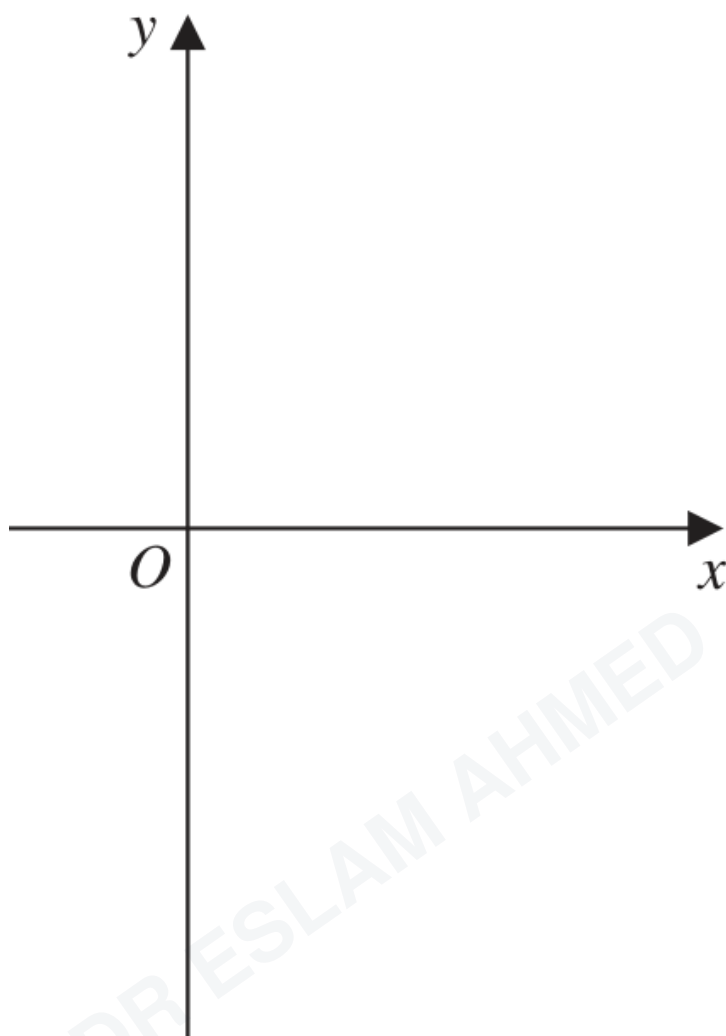
- (c) find the value of k , (2)
- (d) find, using algebra, the coordinates of point Q .

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

QUESTION CONTINUES

Required answer objects follow.

**Diagram 1**

(Total for Question 7 is 10 marks)

Question 6

WMA11/01 January 2023

T07 Transformations of Functions

F02 Transformed algebraic graphs and exact intersections

V02 Translated reciprocal graph followed by line intersection

1

Sketch $y = 6/x$

The graph is a rectangular hyperbola with asymptotes $x = 0$ and $y = 0$, with branches in quadrants I and III.

2

Describe the transformation

$y = \frac{6}{x-2}$ is obtained from $y = \frac{6}{x}$ by a translation 2 units in the positive x-direction.

3

Use P to find k

At $x = -4$ on $y = \frac{6}{x-2}$, $y = \frac{6}{-6} = -1$.

4

Substitute into the line

Since P lies on $y = kx + 7$, $-1 = -4k + 7$, so $k = 2$.

5

Find the second intersection

Solve $\frac{6}{x-2} = 2x + 7$. Multiplying by $x - 2$ gives $6 = (2x + 7)(x - 2)$.

6

Factor and choose Q

$6 = 2x^2 + 3x - 14$, so $2x^2 + 3x - 20 = 0 = (2x - 5)(x + 4)$. The root $x = -4$ is P, so Q has $x = \frac{5}{2}$.

7

Find y for Q

$$y = 2\left(\frac{5}{2}\right) + 7 = 12, \text{ so } Q = \left(\frac{5}{2}, 12\right).$$

FINAL ANSWER

(b) **Translation 2 units right.** (c) $k = 2$. (d) $Q = \left(\frac{5}{2}, 12\right)$.

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Transformations on trigonometric graphs

WHAT THIS FAMILY ASKS

Recognise a basic sinusoidal graph and map a labelled point under phase translation and output reflection/scaling.

VARIANT ROUTE

V01 Identify a sine graph and transform a labelled extremum

1 Complete Classified questions

6 total marks

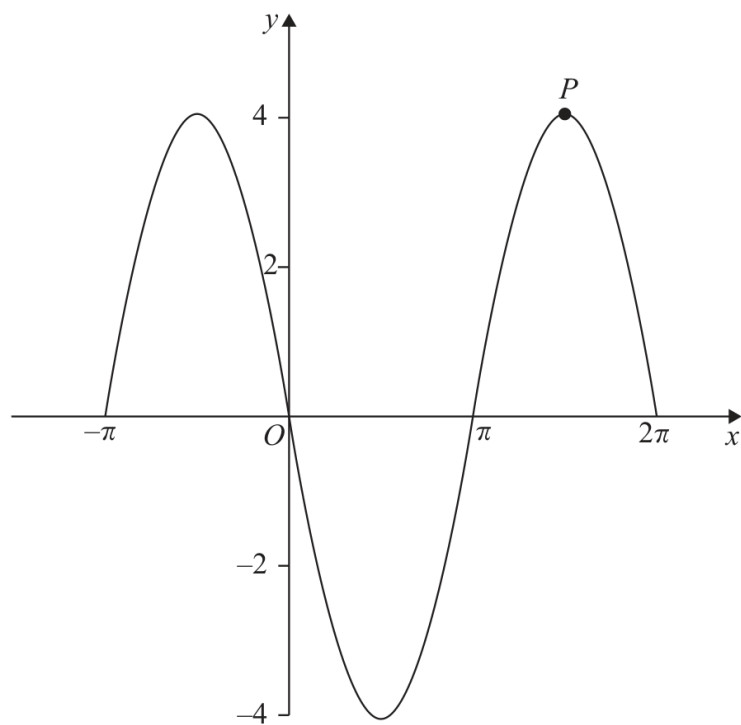


Figure 3

Figure 3 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$

- (a) Write down an expression for $f(x)$

(2)

The point P lies on $y = f(x)$ and is shown in Figure 3.

- (b) State the coordinates of the point to which P is transformed when the graph of $y = f(x)$ is transformed to the graph with equation

(i) $y = f\left(x - \frac{\pi}{6}\right)$

(2)

(ii) $y = -\frac{1}{2}f(x)$

(2)

(Total for Question 9 is 6 marks)

Question 7

WMA11/01 October 2025

T07 Transformations of Functions

F03 Transformations of trigonometric graphs

on V01 Identify a sine graph and transform a labelled extremum

1

Identify amplitude and direction

The graph has amplitude 4, crosses the origin, and initially moves downwards, so $f(x) = -4 \sin x$.

2

Identify P

The labelled maximum is at $P = \left(\frac{3\pi}{2}, 4\right)$.

3

Translate the graph right

$y = f\left(x - \frac{\pi}{6}\right)$ translates the graph right by $\frac{\pi}{6}$.

4

New coordinates for part b(i)

P becomes $\left(\frac{3\pi}{2} + \frac{\pi}{6}, 4\right) = \left(\frac{5\pi}{3}, 4\right)$.

5

Transform y equals negative half f(x)

$y = -\frac{1}{2}f(x)$ leaves x-coordinates unchanged and multiplies y-coordinates by $-\frac{1}{2}$.

6

New coordinates for part b(ii)

P becomes $\left(\frac{3\pi}{2}, -2\right)$.

FINAL ANSWER

$$(a) f(x) = -4 \sin x. \quad (b)(i) \left(\frac{5\pi}{3}, 4 \right). \quad (b)(ii) \left(\frac{3\pi}{2}, -2 \right).$$

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